Live-in-Labs®

Co-Design Report

Submission for Live-in-Labs II

JULY - 2025

Team 7: Abhyudaya
Amrita Vishwa Vidyapeetham, School Of Engineering, Amritapuri Campus

TABLE OF CONTENTS

1. INTRODUCTION

Team Name : Abhyudaya
Team number : 7

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Village / State :
Naala
Uttarakhand

Ward Name: Vishwanath
Ward Member: Matvar Singh Bhandaari (in position for 4 months)

Zonal Coordinator : Indradeo Singh
Contact : 9304626482
Village Coordinator : Shashi Devi

Dates of Village Visit :
14 July 2025 – 19 July 2025 (6 days in Village excluding Travel)

ABOUT THE VILLAGE

Naala is a picturesque village situated in the Ukhimath block and tehsil of Rudraprayag district, Uttarakhand. Positioned along the Kedarnath road, it lies approximately 17 kilometres from the sub-district headquarters in Ukhimath and 44 kilometres from the district headquarters in Rudraprayag. The renowned Kedarnath shrine is located about 46 kilometers away. The village spans an area of 181.5 hectares, characterized by terraced fields cultivating paddy and wheat, reflecting the region's agrarian lifestyle.

According to the 2011 Census, Naala has a population of 1,189 individuals residing in 257 households. The gender distribution includes 600 males and 589 females, resulting in a sex ratio of 982 females per 1,000 males, which is higher than the Uttarakhand state average of 963. Children aged 0–6 years constitute 11% of the population, with a child sex ratio of 914.

The village boasts a commendable literacy rate of 83.89%, surpassing the state average of 78.82%. Male literacy stands at 96.6%, while female literacy is at 71.05%, indicating a significant gender gap that highlights the need for continued efforts in female education.

Naala is deeply rooted in spiritual traditions. At the heart of the village stands the ancient Maa Tripura Sundari temple, believed to be over 900 years old. This temple is a central place of worship and holds immense significance for the villagers, reflecting their deep religious sentiments.

Agriculture and cattle rearing form the backbone of livelihoods in the village. Women primarily engage in farming and animal husbandry, while many men migrate seasonally to Kedarnath to work as drivers, porters, shopkeepers, or tent operators during the pilgrimage season. This seasonal employment brings essential income to many households.

Families live in close-knit clusters, maintaining traditional practices and community ties. The social fabric of the village also reflects a subtle degree of caste-based segregation, with households from different caste groups often residing in separate areas.

From the very first visit, Naala presented itself as a serene and beautiful village. With breathtaking views of the snow-capped Himalayas in the distance and peaceful surroundings, it felt spiritually rich and deeply connected to nature. The Kedarnath Yatra holds immense significance here, bringing a rhythm to village life during its six-month season. The people of Nala are religious, humble, and welcoming.

2. COMMUNITY CHALLENGES

2.1 Identified Problem: Lack of Safe Drinking Water

Naala, a hilly village in Uttarakhand with 257 households, faces a persistent challenge of accessing safe drinking water. Although the village is equipped with a community tank and piped water connections extending to most households, the water is not fit for consumption. Field observations and water quality assessments revealed biological contamination due to poor maintenance of the tank, leakage in distribution pipelines, and absence of treatment facilities. As a result, families avoid using piped water for drinking and instead depend on the dhara (natural spring) located uphill. This reliance creates significant physical strain, especially for women and children who are primarily responsible for water collection. During the rainy season, the steep terrain becomes dangerous, adding to the risks. The problem extends beyond access, as the quality of water consumed directly affects the health and well-being of the community.

2.2 Validation of the Problem

The lack of safe drinking water was validated using multiple methods during the first phase of Live-in-Labs. Participatory Rural Appraisal (PRA) activities such as transect walks, problem tree analysis, and focus group discussions consistently highlighted unsafe water as the community's most pressing concern. Household interviews revealed that almost all families reported foul-smelling, visibly contaminated water from the pipes, making it unsuitable for drinking. Observations during resource mapping showed that the community tank lacked proper cleaning and that pipelines were prone to leakages.

Health data from the Primary Health Centre (PHC) in Guptakashi further confirmed the impact of poor water quality, with 20–25 cases of waterborne illnesses such as typhoid and diarrhoea being reported every season. Focus groups with women emphasized the burden of repeated trips to the dhara, while cultural discussions revealed that villagers held deep faith in the natural purity of spring water, leading to limited awareness about biological contamination and reduced adoption of safe practices like boiling water. Collectively, these data points validated that the problem was both real and recurring, directly affecting health, time, and financial stability.

2.3 Problem Statement

In Naala village, Uttarakhand, households in the lower region are unable to use piped water for drinking due to visible contamination and foul smell, caused by the absence of proper maintenance of the community tank, forcing them to walk long distances to collect water directly from the dhara (natural spring) without boiling, leading to daily physical strain and increased risk of waterborne illnesses

District Level Water Quality Testing and Monitoring Laboratory
Uttarakhand Jal Sansthan, Rudraprayag
Ph (O): 01364-233226, E-mail: ee.ujs.rpg@gmail.com

TC-13230

ULR No.:TC1323025000000162F
Dated: 06/06/2025

Test Report ID: 302/UJS/DLWQTMLR/2025-26

TEST REPORT

Name and Address of Organization/ Customer	**Amrita Vishwa Vidyapeetham, Amritapuri Campus, Kollam, Kerla.**		
Product Description:	Drinking Water	**Sample Description**	One Water sample received from Nala Tap Water, Guptakashi, Rudraprayag.
Discipline / Group:	Chemical/Water		
Reference Sample Identification:	DLWQTML/UJS/2025/June-02		Date of Start of Analysis: 05/06/2025
			Date of End of Analysis : 06/06/2025
Date and Time of Sample Received: 05-06-2025 & 10:20AM		**Sample Collected by:**	Organisation

S.N	Name of Parameters	Results	Desirable Limit (BIS 10500:2012)	Permissible Limit (BIS 10500:2012) in the Absence of alternative Source	Std. Test Method
1.	Colour, Hazen, *Max*	Less than 1	5	15	IS 3025 (part 4)
2.	pH at 25^0 C	7.24	6.5 to 8.5	No Relaxation	IS 3025 (Part-11)
3.	Turbidity, NTU, *Max*	Less than 0.5	1	5	IS 3025 (Part-10)
4.	Total Dissolved Solids (TDS), mg/L, *Max*	48.0	500	2000	IS 3025 (Part-16)
5.	Chloride, mg/L, *Max*	6.0	250	1000	IS 3025 (Part-32)
6.	Total Hardness (as $CaCO_3$), mg/L, *Max*	26.0	200	600	IS 3025(Part 21)
7.	Calcium (as Ca), mg/L, *Max*	6.4	75	200	IS 3025 (Part-40)
8.	Magnesium (as Mg), mg/L, *Max*	Less than 5	30	100	APHA 24th Ed.-3500-Mg B
9.	Total Alkalinity (as $CaCO_3$), mg/L, *Max*	24.0	200	600	IS 3025 (Part 23)

1. The results listed refer only to tested sample and applicable parameters.
2. This certificate is not be reproduced wholly or in parts promotional publicity or legal purpose without permission of authority of testing laboratory.
3. The samples will be destroyed after Seven days from the date of issue of test certificate, unless otherwise agreed with customer.
4. The results apply to the sample as received
5. The Laboratory is responsible for all the information provided in the report, except when information is provided by the customer

**** End of Report****

District Level Water Quality Testing and Monitoring Laboratory Uttarakhand Jal Sansthan, Rudraprayag
Ph (O): 01364-233226, E-mail: ee.ujs.rpg@gmail.com

Test Report ID: 162/UJS/DLWQTMLR/2025-26
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S.N	Name of Parameters	Results	Desirable Limit (BIS 10500:2012)	Permissible Limit (BIS 10500:2012) in the Absence of alternative Source	Std. Test Method
1.	Residual free Chlorine, (mg/l)	Less than 0.2	0.2	1.0	APHA 23rd Ed.-4500-Cl-G
2.	Total Coliform, MPN	Present	Absent	Absent	APHA 23rd Ed.-9223¯B
3.	E. Coliform, MPN	Present	Absent	Absent	APHA 23rd Ed.-9223¯B

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**** End of Report****

Figure 1- Water test Results of Tap water in Naala

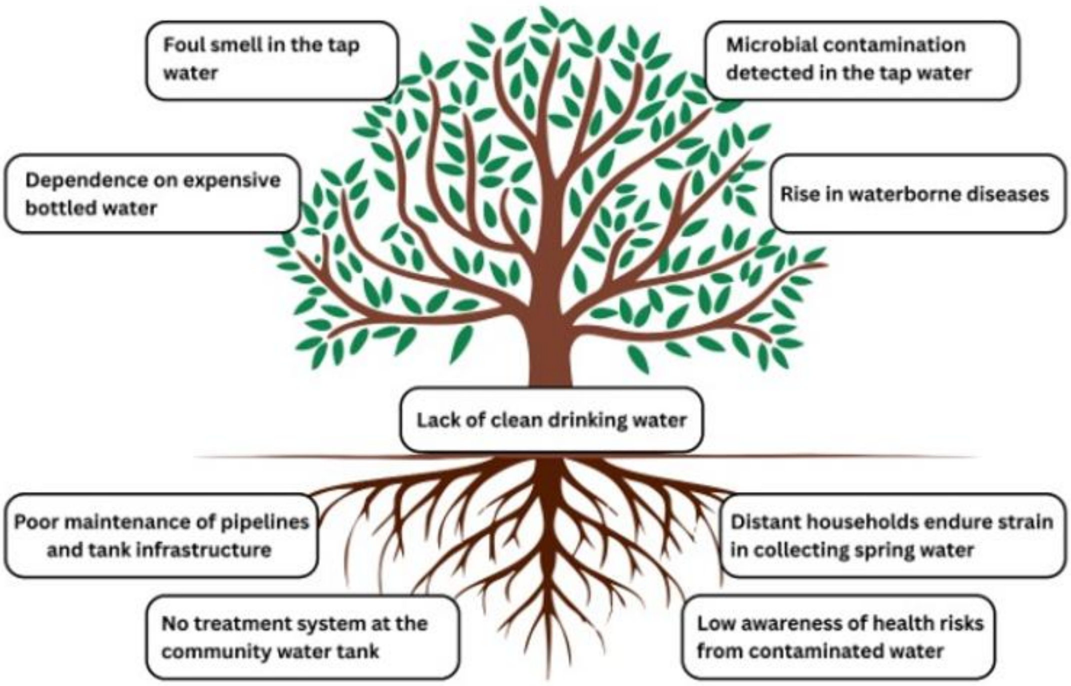
The water test results revealed the presence of E. coli and total coliform bacteria, indicating biological contamination in the sample. The detection of coliforms suggests that the water may have been exposed to sources of microbial pollution, making it unsafe for direct consumption. These findings highlight the need for proper treatment and purification measures to ensure safe drinking water.

3. CONTEXT AND ANALYSIS

3.1 Problem Tree and Causal Factors

The problem tree created during the first Live-in-Labs visit clearly demonstrated that the core issue is the lack of safe drinking water, despite the presence of basic infrastructure. At the root of the tree were causal factors including:

- Direct causes: Poorly maintained storage tanks, leakages in pipelines, and the absence of any treatment or filtration system.
- Indirect causes: Low awareness about safe water practices, reliance on traditional beliefs in the purity of spring water, irregular maintenance schedules, and limited accountability for upkeep.



3.2 Existing Water Infrastructure and Gaps

Naala is comparatively well-equipped in terms of infrastructure when compared to many other rural villages in India. The community has a large water storage tank and piped connections extending to nearly every household. In principle, this infrastructure should ensure equitable access to water. However, the gaps lie in quality and upkeep:

- The storage tank is rarely cleaned and often contains visible silt, algae, and organic matter.
- Distribution pipelines are old and subject to leaks, which allows contaminants to enter the system.

- No proper maintenance and cleaning at the tank, leaving water biologically unsafe.
- Households lack affordable alternatives for purification at the point of use.

Thus, the infrastructure exists but fails to provide its intended benefit - safe drinking water.

Cultural and social factors play a major role in shaping water access and use in Naala. The villagers have a strong cultural and emotional attachment to their water sources, especially the dhara, which is regarded as naturally pure and sacred. This reverence makes it difficult to convince people of the possibility of invisible biological contamination, creating resistance to adopting boiling or filtering practices.

3.3 Impact on Health and Daily Life

The lack of safe water directly impacts the daily lives of Naala's residents:

- Health impacts: Recurring cases of typhoid, diarrhoea, and other waterborne illnesses reported by the Primary Health Centre (20–25 cases per season). Children and the elderly are particularly vulnerable.
- Physical burden: Women and children make multiple trips uphill every day to fetch water from the dhara. This creates significant physical strain and consumes time that could otherwise be used for education or livelihood activities.
- Safety risks: Walking uphill with buckets becomes dangerous in the rainy season due to slippery terrain and landslide risks.

4. EXPLORING SOLUTIONS

4.1 Review of Existing Attempts and Interventions

During the initial field engagement, it became clear that the issue was not the absence of water infrastructure but its poor maintenance and underutilization. Naala village already had a piped water system connected to a community water tank, as well as traditional spring sources. However, contamination occurred due to:

- Poor maintenance of the storage tank, leading to biological growth.
- Leakages in pipelines, which allowed soil and dirty water intrusion.
- Household-level practices where safe water handling was not followed.

Previous interventions, such as piped connections and tank storage, failed to guarantee safe drinking water. Villagers also expressed cultural confidence in the purity of natural spring water, which made awareness efforts challenging. Thus, existing infrastructure could not be fully relied upon for ensuring safe water access.

4.2 Community Aspirations (Problem-Free State)

Through discussions, PRA tools, and participatory storytelling, the aspirations of the community were clarified. Villagers expressed that a "problem-free state" for them would be:

- Reliable access to safe drinking water for all households.
- A common filtration unit located in a place accessible to women, children, and elderly people.
- Culturally respectful infrastructure – addressing caste dynamics through multiple access points or pipes, and incorporating religious or symbolic imagery.
- Ease of use : minimal manual effort, quick collection, and low maintenance.

4.3 Solution Matrix

Problem Situation: Lack of clean drinking water

	Direct	Indirect
High impact	1. Community Water Purifier (Jeevamritham) 2. Community Rainwater Harvesting (7000L Tank) 3. Basic Endpoint Filters	1. Clean & Maintain Community Tank 2.Awareness on Water Treatment
Low impact	1. Rooftop Rainwater Harvesting for 30 Houses *(costly per household)*	1.Demonstration of Traditional Water Purification Methods

4.4 Identifying Solutions

After analyzing the existing challenges, reviewing possible interventions, and validating community needs, the solution identified was the implementation of the Jivamritam project in Naala village. The Jivamritam system was chosen because it provides the right balance of technical effectiveness, cultural alignment, and sustainability. Water test reports from the village confirmed that there is no physical or chemical contamination but biological contamination is present. This meant that advanced treatment technologies like RO were unnecessary, and instead, a simpler multi-stage system was sufficient. Accordingly, the design specifications of the unit included a pre-sediment filter, micron filter, and UV filtration stage to effectively remove biological impurities.

The system is designed to be community-managed, requiring only basic caretaker training, and is affordable and low-maintenance compared to more complex systems. It also ensures inclusivity by addressing cultural sensitivities, such as multiple taps for different groups, which supports greater acceptance

The first level design specifications for the filtration solution are :

- **Filtration Technology**:
  - Pre-sediment filter to remove larger particles.
  - Micron filter for finer dust and impurities.
  - UV filter to kill biological contaminants (bacteria, viruses).
- **Capacity**:
  - Storage tank of 1000 litres, sourced from the dhara.
  - Daily refill cycle based on community's water demand.

4.5 Risk Assessment

While the Jivamritam filtration unit offers a sustainable and community-managed solution, it is important to anticipate potential risks that may arise during its operation and long-term use. Risk assessment helps ensure that both the technical and social aspects of the solution are managed effectively. The table below highlights the key risks identified, their likelihood, and proposed mitigation strategies.

Risk	Likelihood	Mitigation Strategy
1. Villagers stop using the system after 1 month	Medium	Identify peer champions, conduct hands-on demos, and carry out regular follow-ups.
2. Tank is not cleaned regularly	High	Assign cleaning duty to a 3-member local team with a monthly schedule.
3. Water quality degrades due to algae/mosquitoes	Medium	Install tank lid, mesh screens, and raise awareness on proper closure and use.
4. Monsoon rainfall is lower than expected	Low	Emphasize this as a supplementary source and maintain other backup options.
5. Community loses interest post-handover	High	Plan phased handover with a 3-month support period and periodic engagement sessions.

5. CO-DESIGN PROCESS

5.1 Approach: Human-Centered Design (HCD) Tools

The co-design process for the Jivamritam project in Naala was anchored in the Human-Centered Design (HCD) approach, which places the community at the heart of problem-solving. Unlike top-down interventions that often fail due to lack of cultural fit or ownership, HCD emphasizes empathy, participation, and iterative feedback, ensuring that solutions are designed with the community, not for them.

Why HCD?

1. **Community Ownership and Acceptance:** By involving villagers in every step from identifying problems to finalizing the filtration site and caretaker, HCD ensured that the solution aligned with their cultural values and daily realities.
2. **Context-Specific Insights:** Villagers provided knowledge that no external survey could capture (e.g., caste sensitivities requiring multiple taps, cultural symbols for acceptance, local practices of water use).
3. **Behavioural Change:** Through interactive tools like storytelling, poster cards, and games, HCD helped overcome cultural resistance to the idea of invisible contamination in spring water.
4. **Sustainability:** Solutions co-created with the community are more likely to be maintained, as people feel a sense of responsibility and pride in the project.
5. **Inclusive Decision-Making:** Tools like group sketching and participatory games allowed voices from women, children, and marginalized groups to be equally represented.

5.2 Tools and Activities Planned

To collect and verify missing information with the community, a set of Human-Centered Design (HCD) tools were implemented. Each tool was chosen for its ability to engage villagers interactively, avoid direct questioning that might cause resistance, and allow for consensus-building. The strategy ensured that villagers not only shared their perspectives but also actively participated in shaping the design of the filtration system.

.1. Location Selection Game – “Stand by Choice”

- Objective: To finalize the most suitable location for the filtration unit, balancing accessibility, safety, and cultural considerations.
- What we wanted to find out: Community preferences on location attributes such as proximity to source water, availability of electricity, safety, cleanliness, and ease of access for women and children.
- Three potential sites should be identified by the village Pradhan will be marked on the ground as squares. Villagers will physically stood in the square representing their preferred option for each attribute. After discussions, they will be allowed to change their choice. This

will create a transparent and participatory decision-making process, leading to consensus on the most appropriate

2. Caretaker Selection – Voting

- Objective: To identify a trustworthy community caretaker for the filtration unit.
- What we wanted to find out: The qualities villagers associate with a good caretaker and who among the nominated individuals they trust most.
- Villagers will be briefed on caretaker responsibilities (daily monitoring, cleaning, managing breakdowns). They then list the key attributes needed. Candidates will get nominated and finally selected through a voting process and discussion , ensuring fairness and community acceptance.

3. Storytelling and Poster Cards – "Invisible Germs"

- Objective: To raise awareness about biological contamination in piped and dhara water, which is invisible but harmful.
- What we wanted to find out: Villagers' perceptions of water safety and openness to adopting safe practices such as filtering or boiling.
- Using poster cards, students will explain how water could be contaminated by animals, storage conditions, and pipeline leaks. Through storytelling, participants have to be engaged in discussing how these germs could lead to illnesses. The session especially targets women, as they are the main water handlers.

4. Timeline Mapping – "Water Clock"

- Objective: To determine the timings when filtered water would be most useful for the community.
- What we wanted to find out: The daily and seasonal water usage patterns, peak demand times, and scheduling feasibility.
- A large clock is drawn on a chart, and villagers will put ticks on the times when they most need filtered water. After group discussion, consensus will be built on fixed operational timings for the unit.

5. Group Sketching – Designing the Filter Surroundings

- Objective: To integrate community preferences into the look and layout of the filtration site, making it culturally acceptable and user-friendly.
- What we wanted to find out: Desired structural features (platforms, dual taps, drainage), protective elements (roof, fencing), and visual aspects (wall art, slogans, religious symbols).

- Villagers can be divided into groups (e.g., higher caste, lower caste, women, children). Each group sketches their ideas on chart paper. These ideas will be discussed collectively, and key suggestions will be noted for incorporation into the final site design. This session can also be used to answer the questions of villagers regarding the solution and clear their doubts.

6. Filter Awareness for Children

- Objective: To create early awareness of safe drinking water practices and encourage children to influence family behaviour.
- What we wanted to find out: Whether children could understand basic filtration concepts and share the knowledge at home.
- A simple prototype bottle filter with gravel, sand, and charcoal layers can be demonstrated. Children has to name each layer and guess its role. The session should be both educational and interactive.

5.3 Day-wise Plan of Activities

To ensure completion of Co-design process and systematic implementation of Jivamritam water filter in Naala during the Live-in-Labs II, a detailed plan of activities was prepared in advance. The following table summarizes the planned day wise activities:

Date/Day	Morning	Break	Afternoon
14/07/2025 - Monday	Reintroduction and problem tree validation with villagers in small groups, Observe the current water infrastructure		Meet Stakeholders - Pradhan, Anganawadi teacher ,Temple priest
15/07/2025-Tuesday	Visit Block Panchayat office authorities and water authorities		Group discussion - design games(Location),poster cards
16/07/2025-Wednesday	Focus group discussion with women - Storytelling		Visit the tuition centre for informing about the demonstration of the prototype , filter awareness activity
17/07/2025- Thursday	Visit PHC doctor to gather insights. Finalize the location, Arrange for prototype demonstration		Demonstration of the filter. Awareness session on filtering/boiling drinking water. Group sketching , roleplay, Election for choosing caretaker
18/07/2025- Friday	Post implementation survey in households. Get feedback		Group discussion with women for finalizing timings - Timeline mapping
19/07/2025- Saturday	Give training for the caretaker , Brainstorming with water committee		Implement improvements in filter based on feedback
20/07/2025-Sunday	Buffer day (if any delays due to seasonal reasons)		

FIELD WORK

6. INNOVATION FINDINGS

The fieldwork for the co-design phase in Naala village began on 14 July 2025. The initial days were dedicated to meeting key stakeholders including the ward member, the village priest, and the village coordinator to reintroduce the project and build trust around the proposed solution. Parallelly, discussions were held with the Jivamritam technical team, who were on-site to oversee the installation of the filtration unit, including site preparation, wiring, water source connection, and system testing.

Once the groundwork was in place, the co-design process with the villagers was initiated. A series of participatory tools and activities were conducted to gather community input, validate decisions, and finalize key aspects of the project such as location of the filtration unit, caretaker selection, operational timings, awareness practices, and beautification of the site. The process ensured that both technical implementation and community engagement moved hand in hand, fostering ownership and inclusivity.

6.1 Detailed Results from Each Tool with Data

The following section details the tools applied, the activities carried out in the field, the results obtained, and supporting evidence from photos, sketches, and community feedback.

1. **Group Discussions**

    - Site Selection

Site selection for the filter was guided by the insights and observations gathered during our Lila 1 visit to Naala village. Based on these observations, we had pre-identified three potential sites for implementing the water filtration system: the Government Public School, Maa Tripurasundari Lalita Mai Temple, and the Anganwadi, considering Accessibility, Safety, Cleanliness, Source Water Proximity, and Electricity Availability. Since our Jivamritam team had reached the village earlier, members Vinod and Mahesh began initial interactions with local residents, the ward member, and community leaders to discuss the project implementation. They also visited the proposed sites to assess the Basic Infrastructure and Service Parameters. From these interactions, the Temple emerged as the most suitable site. The priest (Pandit Vishnu Dutt) agreed to provide the space, electricity was freely available (reducing maintenance costs), the location was central and accessible, and a reliable water source was nearby.

On the first day of Phase 2, we reintroduced ourselves to the villagers, explained the problem–solution approach and implementation plan, and collected their feedback on the preferred site. Concerns were raised regarding the temple site, as it was traditionally restricted to upper-caste residents which limited access for other community members. To address this, we visited the site with the priest (Pandit Vishnu Dutt) and the village coordinator (Shashi Devi ji) to discuss inclusivity and feasibility. We also met with the ward member (Anuj Bandari) to better understand the caste-based segregation issues related to site selection. On the following day, a Focused Group Discussion was conducted with residents from the lower part of the village to gather their views, concerns, and suggestions. After this discussion, villagers agreed to implement the filter at the Temple, with the condition that two separate taps be connected to the storage tank, similar to the traditional Dhara water collection point. This ensured the filtration system was inclusive, socially accepted, and sustainable. Overall, the site selection was successful, balancing technical suitability with community needs to make the water filter would be accessible and beneficial to the entire community.

Figure 2- Maa Tripura Sundari temple

Figure 3- Site selected for Filter

- Water Management Committee

During the implementation of the water filtration system in Naala village, community ownership, long-term maintenance, and sustainable management were prioritized as core objectives. To ensure that the system would be effectively monitored and maintained, a collective brainstorming session was organized with the ward member, villagers, and the Jivamritam team, focusing on responsibilities for upkeep and addressing future costs, which were agreed to be supported by the Panchayat. A village-wide meeting was then convened, where all residents, local leaders, and the ward members participated in discussions about forming a Water Management Committee. Together, villagers co-decided on the criteria for responsible caretakers which are reliability and commitment, availability, and community trustworthiness. Based on these discussions, it was agreed that the committee should have a balanced representation of men and women to ensure fairness and inclusivity.

Through this participatory process, the committee members were collectively nominated and endorsed by the community. The selected members were Shashi Devi, Manorama Devi**,** Saraswathy Devi, Yeshpal Negi, Akhilesh Pharashi, and Mathvar Singh Bhandari**.** To formally acknowledge their roles, signatures were collected from the villagers and ward members, which served as a community vote of confidence in the process. Finally**,** the primary caretaker of the filtration unit was chosen as the village temple priest, Pandit Vishnu Dutt**,** reflecting the trust he holds in the community. Other committee members pledged to support him in operations, maintenance oversight, and ensuring equitable access.

Figure 4- Water Committee Members

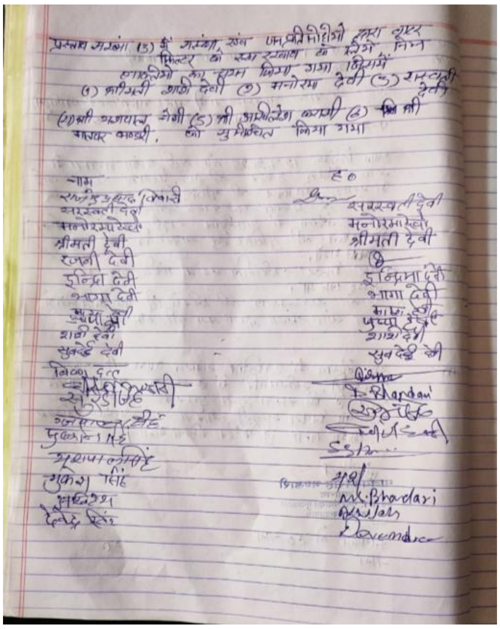


Figure 5- Signatures of Villagers approving Committee

2. **Poster Cards and Storytelling on Biological Contamination**

During group discussions with villagers, poster cards and storytelling sessions were used to raise awareness about the importance of filtration and the risks of biological contamination. The posters illustrated possible sources of contamination such as animals near the stream, unclean storage tanks, and leaking pipelines while the story explained how invisible germs, which have no colour or smell, can enter water and cause illness. Care was taken to explain these concepts sensitively, without offending cultural and religious reverence for the dhara . The focus was placed on the journey of water from the source to the household, highlighting how contamination often occurs along the way rather than at the spring itself.

As a result, many villagers began to recognize that water which appears clear is not always safe for drinking. Women, who are primarily responsible for water collection and storage, acknowledged the role of handling practices in contamination and expressed willingness to adopt safer practices like filtration or boiling.

Figure 6- Story Telling activity

3. **Timeline Mapping**

On the day of the inauguration, a participatory timeline mapping activity was conducted to collectively decide the timings for operating the filtration unit. A large chart with a clock diagram was placed in the village common area, and villagers were invited to mark their preferred hours for collecting water. After discussion, the community collectively agreed on two main slots: 10–11 am in the morning and 4–5 pm in the evening. These timings were not random but strongly connected to the agricultural and household practices of the village. The morning slot was chosen to align with the time when women return from their early agricultural tasks and begin preparing meals, while the evening slot was selected to coincide with the completion of fieldwork and cattle-rearing activities, when families typically require fresh water for both cooking and drinking. This activity allowed villagers to actively co-decide the filtration schedule in a way that fit naturally into their daily routines, ensuring inclusivity and ownership while reinforcing the cultural importance of aligning community resources with agricultural life.

Figure 7- Timeline Mapping Activity

4. Group Sketching

A brainstorming and group sketching session was conducted with two community groups, organized separately for the upper caste and lower caste households to respect existing social divisions. The activity aimed to capture their preferences for the environment around the filter and water taps , including the design of platforms, the type of awareness materials to be displayed, and the overall cleanliness and beautification of the area. Both groups actively participated by sketching and describing their ideas. The upper caste group emphasized the need for a strong, raised platform to place buckets during collection since the water taps are located on a corner in a raised position, along with regular cleaning of the surroundings and posters carrying slogans like “Water is Life”. They also had clear preferences on images that should be on the posters. The lower caste people suggested relocating their water tap to a different location in order to reduce crowding and create more space for people to stand comfortably while collecting water. For awareness materials, they preferred bright, pleasing visuals and easy to understand slogans. Across both groups, hygiene, accessibility, and awareness emerged as common priorities. The session demonstrated that beyond the filter itself, villagers value improvements in the surrounding environment, clear instructions, and visible awareness messages as key factors in ensuring the system’s acceptance and long-term use. All the suggestions were incorporated and implemented. Incorporating community inputs was important as it ensured that the system reflected their real needs, increased their trust and ownership of the project, and encouraged long-term acceptance and use of the filtration unit.

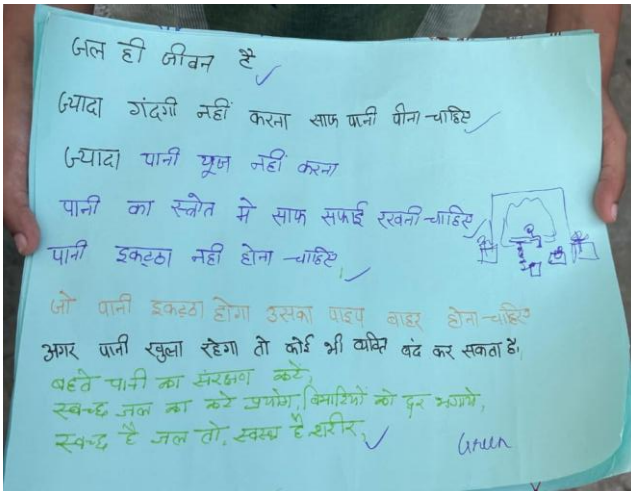


Figure 8- Results of group sketching activity

5. Demonstration

As part of the project, two demonstrations were conducted to familiarize the villagers and the water management committee with the concept and functioning of the filtration system.

The first demonstration was carried out on the inauguration day, where a simple bottle filter prototype was used to show how a water filter works. This was done in front of all the villagers to make the idea of filtration more tangible and easier to understand. The purpose of this demonstration was to build awareness on the importance of clean water, generate curiosity, and create trust in the technology by showing the process in a clear, relatable manner. By visualizing how impurities are removed, villagers could see the direct health benefits of using filtered water, which helped in encouraging its adoption.

Figure 9- Bottle filter Demonstration

The second demonstration involved the actual operation of the installed filtration system. Members of the water management committee were given hands-on training by the Jivamritam team. Each step of the operation from switching on the unit, regulating water flow, cleaning procedures, and safety precautions was explained in detail. Importantly, every committee member was invited to practice these steps themselves under supervision. This ensured that they not only understood the process theoretically but also gained the confidence to independently manage the system. Such practical involvement is critical because it reduces dependency on external teams, ensures proper long-term maintenance, and instils a sense of responsibility and ownership within the committee.

Figure 10- Demonstration of Jivamritam filter

6.2 Summary of Needs Document

The needs identified in the earlier stage were addressed through the interventions and activities undertaken during the second village visit.

1. Filter systems for the pipeline water

    - This was the most urgent need and has been fully met. A community-level filtration system has already been installed. Demonstrations were carried out with villagers to show how filtration works, and the water management committee was trained by the Jivamritam team on the operation and maintenance of the unit. This ensures that safe drinking water is available and that the community has the skills to sustain the system.

2. Maintenance and cleaning of the water tank

    - To address hygiene concerns, a raised platform was constructed for the tank, and the floor around the filter unit was concreted for stability and cleanliness. The water management committee has been assigned the responsibility of cleaning and maintenance, with awareness given on the importance of regular upkeep.

3. Cleaning the downhill flow of the dhara and awareness on waste disposal

    - While direct cleaning of the dhara was not undertaken during this visit, awareness was created among villagers through co-design sessions and posters on the importance of hygiene and protecting water sources from waste. This remains a point for future engagement.

4. Promoting alternative sources like rainwater harvesting

    - Although rainwater harvesting was identified as a long-term priority, it was not implemented in this phase. However, it remains an important strategy to reduce dependence on piped water, especially during peak tourist season. Discussions on this can be continued in subsequent interventions.

5. Addressing caste-based exclusion from community sources

    - While fully resolving this barrier is difficult, progress was made. Separate co-design sessions were held with both caste groups to ensure their voices were equally heard. The lower caste group's request for relocating their pipeline was implemented, along with the construction of platforms and awareness posters. This demonstrated respect for their preferences and helped build trust, making the system more inclusive.

6. Decline in Waterborne Diseases

- The PHC doctor was consulted and agreed to help monitor cases of waterborne illnesses over time. With safe filtered water now available, it is expected that the frequency of illnesses like typhoid and diarrhoea will reduce, and future health records will serve as validation of this improvement.

In summary, the findings and actions from the visit responded to nearly all priority needs. The installation of the filtration system, infrastructural improvements, maintenance planning, and awareness initiatives directly addressed the most critical needs.

6.3 Final Design of the Proposed Solution

The final solution is a community water distribution system with integrated filtration, storage, and user-friendly access, developed based on extensive field feedback. The design ensures equitable access, safety, and community involvement, while also incorporating aesthetic enhancements to encourage proper usage and water conservation.

Water Source & Collection

- The source of water is the piped supply from the temple, which originates from the Jal Jeevan Mission water tank.
- Water is collected in a primary collection tank before filtration.
- It is then passed through the Jivamritam filter, followed by a three-stage filtration system:
    1. Sediment Filter – removes visible particles.
    2. Activated Carbon Filter – eliminates foul smell and improves taste.
    3. UV Sterilization Unit – ensures the water is free from bacteria and pathogens.

Storage & Distribution

- Water is stored in a 1000-litre elevated tank for gravity-fed distribution.
- Two separate distribution pipes have been installed to ensure equitable access:

    One pipe for higher-caste households.

    One pipe for lower-caste households.

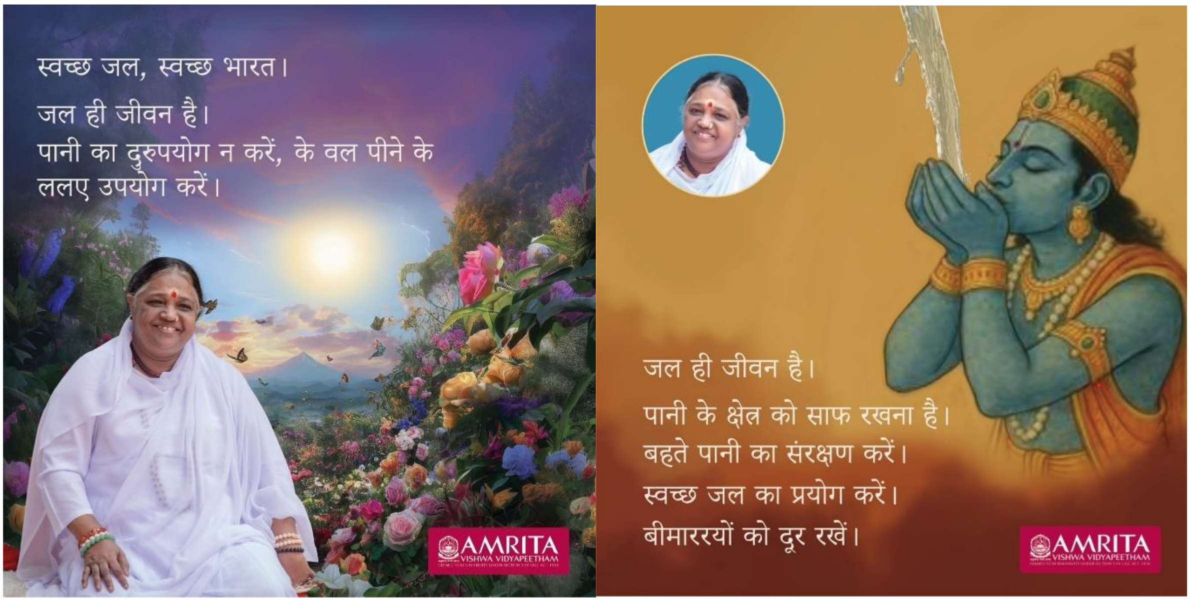
- Flow-regulated taps are installed at child-friendly heights.
- Platforms below the taps have been added to prevent water spillage and facilitate safe usage.

Beautification of the Area

- Raised platforms have been built for the tanks and filtration units to enhance aesthetics and maintain hygiene.
- The community recommended visual elements to promote water awareness:

  Higher-caste households: Laminated posters depicting Lord Vishnu drinking water, Amma image with water awareness slogans, and a ghost (Chudeli) reminding children not to waste water.

  Lower-caste households: Laminated posters with sunrise, flowers, mountains, Amma image, and simple Hindi slogans for easy understanding.
- Both groups advised laminated posters instead of painted ones to prevent damage from rain.
- These posters have been placed near the taps to encourage proper water usage and create a visually appealing environment.
- Group sketching sessions were conducted, and the final design document was shared with the Jivamritam team for implementation.



Posters designed based on community inputs

Maintenance & Community Management

- A local water committee oversees the system, addresses operational issues, and ensures equitable distribution.
- Minimal technical skills are required, making the system easy to maintain by community members.

The final design provides a reliable and safe supply of clean water to all households. It ensures equitable access across different caste groups, addressing social inclusion concerns. The system is easy to maintain and is designed to be safe for children and the elderly. It incorporates community preferences, ownership, and local aesthetics, fostering a sense of participation and pride. The use of visual awareness elements encourages water conservation among community members. Overall, the design is sustainable and low-cost, making it practical for long-term operation.

6.4 Challenges

Main Challenge Faced: Managing Social Inclusion Across Caste Groups

The primary challenge during the fieldwork was bringing together both higher-caste and lower-caste households to collaboratively accept and use the proposed water solution. This was a particularly sensitive issue because the water filtration system was planned near the temple area, a space that lower-caste residents traditionally have restricted access to.

During house visits to lower-caste households, residents were initially apprehensive about the project. They raised concerns about whether they would be allowed to use the water filter freely and voiced fears of potential discrimination or social backlash.

Villagers expressed differing opinions among themselves, highlighting the complexity of caste-based perceptions and expectations. Many were worried about possible conflicts over ownership, access, and fairness if only a single water source was installed.

Actions Taken:

- We carefully managed this sensitive issue by engaging with local governance and community leaders. The Panchayat and ward member were consulted to help mediate the situation, provide assurance, and take responsibility for monitoring access.
- We held discussions with lower-caste residents, explaining clearly that they would face no barriers in using the water supply and encouraging them to share their preferences and concerns.
- Based on these conversations, the design was modified to include two separate taps, one for higher-caste households and one for lower-caste households (original plan was one tap for all) The taps were strategically located away from the temple area to respect cultural sensitivities and ensure comfort and confidence in usage.
- The design decision also considered visibility and accessibility, making it easier for children, elderly, and all households to use the system without feeling excluded.

Importance of the Approach:

This approach helped build trust and acceptance among all community members, reducing the risk of social tension and conflict. It emphasized the value of participatory design, showing that community input can lead to practical improvements such as separate taps that were not part of the original plan. By addressing these social and cultural sensitivities, the project became inclusive, socially acceptable, and sustainable, ensuring equitable access to clean water for all households. Moreover, this process provided insights into community dynamics and local governance support, which will be crucial for the ongoing maintenance, ownership, and long-term success of the system.

7. IMPLEMENTATION

The implementation of the water filtration system was carried out in close collaboration with the Jivamritam team and active participation from the community. The process involved setting up the filtration unit, electrical works, and plumbing, with villagers assisting in moving materials, arranging the filter components, and completing on-site tasks. The pre-sediment filter was carefully filled and packed with the filter media, and the team also tested the UV unit and pumping system to ensure proper functioning. The entire setup process took three days before the formal inauguration. The filtration system has a storage capacity of 1000 litres and an outflow rate of 500 litres per hour, ensuring a continuous and sufficient supply of clean water to meet the daily needs of the community efficiently and reliably. This capacity and flow rate balance the demand from village members and practical considerations of local maintenance and operation. Over a period of 4 hours, the filter system fulfils the community's drinking water requirement of approximately 2100 litres for about 700 residents, based on an estimated daily need per person. This filter design addresses both the technical and social dimensions of water filtration in a rural Himalayan context, ensuring safety, functionality, and local ownership for long-term success

The filtration unit was inaugurated on 18th July 2025 with a traditional lamp lighting and ribbon-cutting ceremony, attended by community members. During the inauguration, a brief demonstration of the filter’s working process was provided, highlighting all components and explaining usage instructions to villagers. Beautification and cleaning works around the tanks, taps, and surrounding area were carried out in the following days.

On the last day of the visit, the filter became fully operational, and clean water was made available to all households. The inauguration included the priest personally drinking water from the filter to endorse its safety. Children were invited to fill their water bottles and provide feedback on the water quality. The response was overwhelmingly positive, with villagers appreciating the taste, clarity, and safety of the filtered water, confirming the successful implementation of the system.

Figure 11- Inauguration Ceremony

Figure 12- Jivamritam Filter implemented in Naala

Awareness Sessions for Women and Children

As part of the project, awareness sessions were conducted for both women and children to educate them on safe water practices. For women, these sessions took place during focus group discussions, where they were engaged in conversations about the importance of boiling and filtering water before consumption and the health risks associated with contaminated water. For children, interactive sessions were held at the tuition centre, using animated videos and visual demonstrations to make the learning engaging and easy to understand. The sessions emphasized the diseases caused by unsafe water, proper hygiene practices, and the benefits of using the newly installed filtration system. These activities not only increased knowledge but also encouraged behavioural changes, promoting safe water usage and hygiene within the community.

Figure 13- Awareness Session for Children

Figure 14- Session using Poster Cards

Community Engagement

The successful implementation of the water filtration system in Naala village was made possible through the wholehearted involvement of the villagers. From the very beginning, the community expressed their willingness to support the project in every possible way. This spirit of cooperation became the foundation of the project's success. Their active participation was visible in real, everyday scenarios. For instance, when we visited the temple site chosen for the filtration unit, the area was overgrown with plants, but without hesitation, the temple priest began cleaning the entire space himself, setting an inspiring example for others. During the actual implementation at the filter site, villagers would often sit nearby and extend help wherever possible, whether by moving materials, arranging tools, or simply offering cool drinks to us as we worked. One day, after we fixed the location for constructing the water collecting point, the place was covered with plants and stones. As we started cleaning, a woman on her way to the fields stopped, joined us, and helped in clearing the site, ensuring the site remained neat and functional. Her small act reflected the villagers' readiness to contribute to the project in any way they could.

During discussions and brainstorming sessions, there was active participation from the villagers. They shared their perspectives openly, which helped us cross-check and refine our decisions. In addition to discussions, they actively contributed to planning and creative activities. For example, they actively engaged in timeline mapping to decide operational hours for the filter, ensuring the schedule suited the entire community. Similarly, they participated in group sketching activities suggesting wall art, awareness slogans, structural improvements like platforms and pipeline adjustments, and beautification ideas to make the filtration site both functional and welcoming. Their inputs in both decision-making and design reflected a strong sense of ownership and responsibility. Their support not only ensured the smooth execution of the project but also laid the foundation for its long-term sustainability, driven by the pride and confidence of the people of Naala village.

Figure 15- Team Abhyudaya with Villagers

8. CONCLUSION

8.1 Summary of Outcomes

The project successfully implemented a community water distribution and filtration system that provides safe, reliable, and equitable access to clean drinking water for all households. Through extensive fieldwork and co-design, the system was tailored to community preferences and social sensitivities, ensuring participation from both higher-caste and lower-caste households. The system integrates multi-stage filtration (pre-sediment, Jivamritam, activated carbon, UV), a 1000-litre elevated storage tank, and two separate taps to guarantee equitable access. Visual awareness elements and raised platforms enhance usability, hygiene, and water conservation education. Overall, the project strengthened community ownership, inclusivity, and sustainability while addressing both technical and social challenges.

8.2 Recommendations for the Community

- Follow a monthly cleaning schedule for the storage tank, filters, and taps to ensure continued hygiene and optimal functioning.
- Encourage children and households to actively use the water responsibly, guided by the awareness posters and community education initiatives.
- Monitor the functioning of the UV unit and filter media, replacing components as recommended to maintain water safety.
- Continue community engagement and feedback sessions to address any emerging concerns and strengthen collective ownership.

8.3 Replication Potential in Other Villages

The project demonstrates a replicable model for other villages facing similar challenges in access to safe water. Key factors that support replication include:

- Use of piped water or local natural sources combined with multi-stage filtration systems suitable for varying capacities.
- Participatory design approach, ensuring community preferences, cultural sensitivities, and social equity are integrated into the solution.
- Simple construction and maintenance requirements, allowing communities to manage the system with minimal technical expertise.
- Integration of visual awareness elements and infrastructure design that encourages proper usage and hygiene.
- The approach highlights that sensitive social management and local governance support are crucial for acceptance and sustainability, making it suitable for scaling in other rural contexts.

9. ACKNOWLEDGEMENT

On behalf of Team ABHYUDAYA, we express our sincere gratitude to Sri Mata Amritanandamayi Devi, Chancellor, Amrita Vishwa Vidyapeetham, for her visionary guidance, which continues to inspire the team to pursue selfless service to society. We also extend our heartfelt gratitude to the management of Amrita Vishwa Vidyapeetham, Amrita School of Sustainable Development, for introducing us to the Live-in-Labs course and providing us with the invaluable opportunity to participate. We are deeply thankful to the Live-in-Labs team for their planning of the village visit, ensuring every aspect of our travel was flawlessly executed, leading to a seamless experience. We extend our heartfelt gratitude to our zonal coordinator, Mr. Indradeo Singh for his consistent guidance and support throughout the duration of our village engagement. His insights and timely assistance played a crucial role in ensuring the smooth execution of our field activities. We are equally thankful to our village coordinator, Shashidevi whose active involvement and local knowledge greatly facilitated our interactions with the community and helped us gain a deeper understanding of the village context. We also appreciate the warmth and hospitality extended to us by the villagers, who welcomed us into their spaces and shared their experiences with openness and patience. Their cooperation was invaluable in helping us collect meaningful data.

Most importantly, we would like to sincerely acknowledge the invaluable support of the Jivamritam team throughout the project and field visit. In particular, Mahesh Ji from the Jivamritam team and Vinod Ji from Ammachi Labs played a key role in the implementation of the filtration system and in facilitating the co-design process with the community. Their guidance was crucial in carrying out the practical setup of the filtration unit, teaching villagers how to operate and maintain it, and providing insights during discussions that helped refine the design and address challenges effectively. Their technical and social perspectives greatly enriched both the planning and execution phases. We are deeply grateful for their time, expertise, and dedication, which not only made the project successful but also enhanced the theoretical understanding and approach of the project team.

10. ANNEXES – PHOTOS


GPS Map Camera
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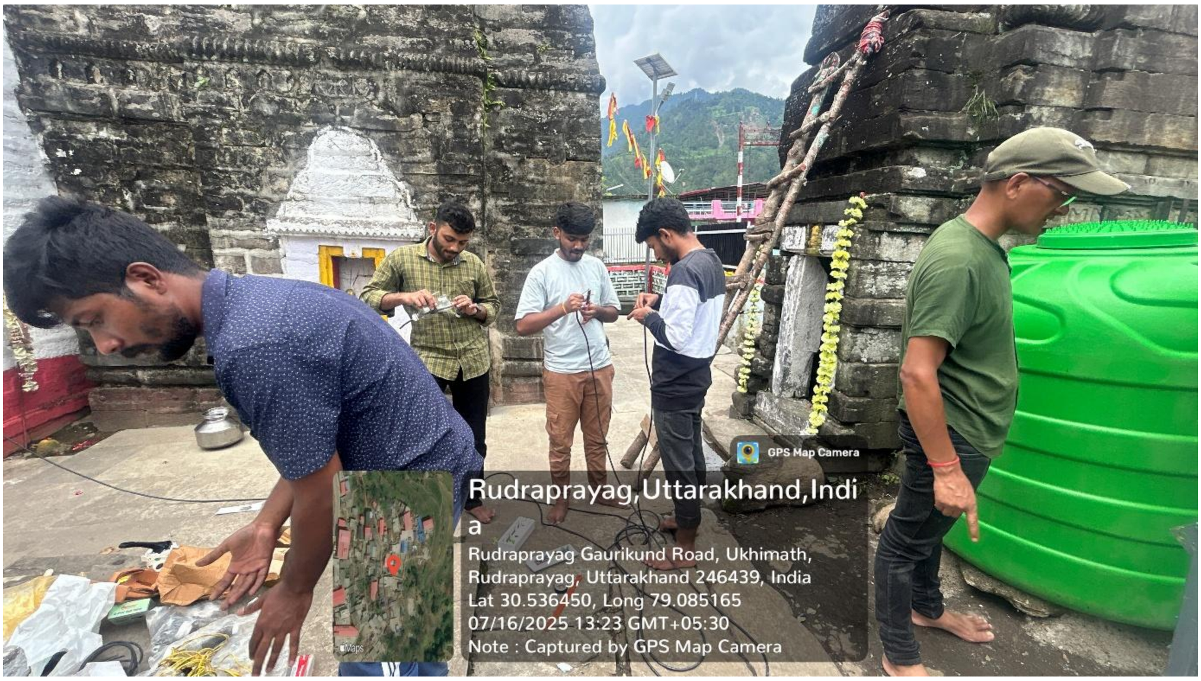
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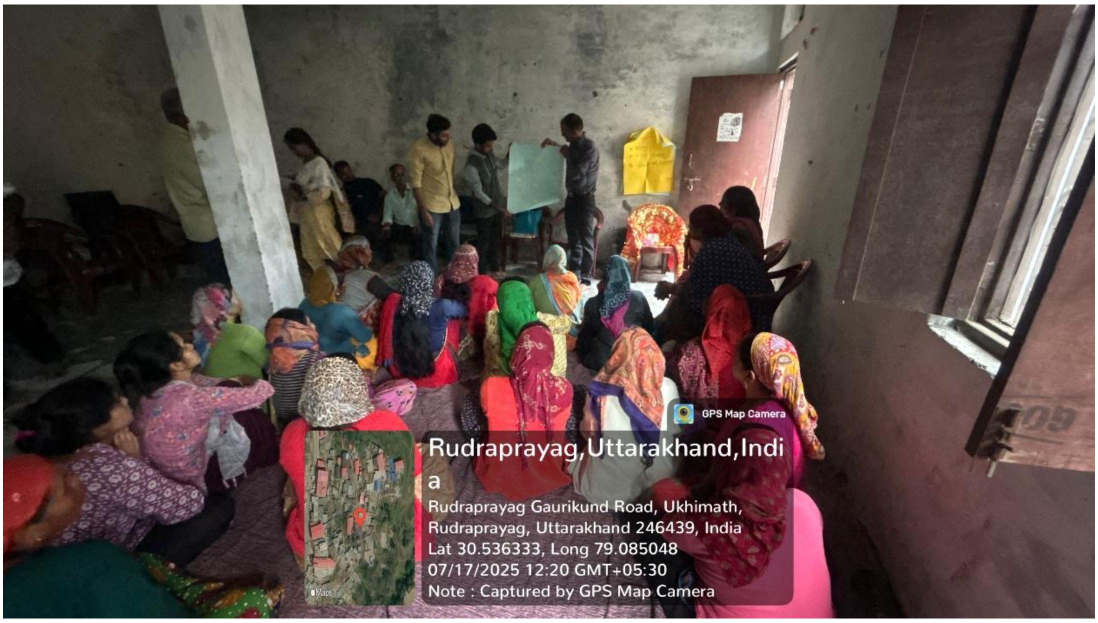
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